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A review of the principles and applications of thermal control

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ABSTRACT

This is a brief introduction to the control of thermal systems and some of its applications. Fundamentals of the linear theory applied to ordinary and partial differential equations are discussed from the point of view of system and output controllability. Some nonlinear effects that are characteristic of thermal systems are pointed out, and commonly used system identification models presented. Details of the control of lumped convective systems with on-off and PID control are given. Conductive, convective and advective phenomena are studied using one-dimensional models in the context of fins, long ducts and heat exchangers. Finite-difference approximations are employed to enable the use of finite-dimensional theory. Applications to complex thermal systems like heat exchangers and networks are also described and representative experimental results shown. An extensive list of references for further study is included.

1 INTRODUCTION

There are many kinds of thermal systems in common industrial, transportation and domestic use that need to be controlled in some manner, and there are many ways in which that can be done. One can give the example of heat exchangers [1,2], environmental control in buildings [3-8], satellites [9-12], thermal packaging of electronic components [13, 14], manufacturing [15], rapid thermal processing of computer chips [16-18], and many others. If precise control is not required, or if the process is very slow, control may simply be *manual*; otherwise some sort of mechanical or electrical feedback system has to be put in place for it to be *automatic*.

Most thermal systems are generally *complex* involving diverse physical processes. These include natural and forced convection, radiation, complex geometries, property variation with temperature, nonlinearities and bifurcations, hydrodynamic instability, turbulence, multi-phase flows, or chemical reaction. It is common to have large uncertainties in the values of heat transfer coefficients, approximations due to using lumped parameters instead of distributed temperature fields, or material properties that may not be accurately known. In this context, a complex system can be defined as one that is made up of sub-systems, each one of which can be analyzed and computed, but when put together presents such a massive computational problem so as to be practically intractable. For this reason large, commonly used engineering systems are hard to model exactly from first principles, and even when this is possible the dynamic responses of the models are impossible to determine computationally in real time. Most often some degree of approximation has to be made to the mathematical model. Approximate correlations from empirical data are also heavily used in practice. The two major reasons for which control systems are needed to enable a thermal system to function as desired are the approximations used

during design and the existence of unpredictable external and internal disturbances which were not taken into account.

There are many aspects of thermal control that will not be treated in this brief review. The most important of these are hardware considerations; all kinds of sensors and actuators [19, 20] developed for measurement and actuation are used in the control of thermal systems. Many controllers are also computer based, and the use of microprocessors [21,22] and PCs in machines, devices and plants is commonplace. Flow control, which is closely related to and is often an integral part of thermal control, has its own extensive literature [23]. Discrete-time (as opposed to continuous-time) systems will not be described. The present paper will, however, concentrate only on the basic principles relating to the theory of control as applied to thermal problems, but even then it will be impossible to go into any depth within the space available. This is only an introduction, and the interested reader should look at the literature that is cited for further details. There are good texts and monographs available on the basics of control theory [24-27], process control [28-31], nonlinear control [32], infinite-dimensional systems [33, 34], and mathematics of control [35, 36] that can be consulted. These are all topics that include and are included within thermal control.

2 SYSTEMS

Some basic ideas of systems will be defined here even though, because of the generality involved, it is hard to be specific at this stage.

2.1 Systems without control

The dynamic behavior of any thermal system (often called the open-loop system or *plant* to distinguish it from the system with controller described below), schematically shown in Fig. 1(a), may be mathematically represented as

$$\mathcal{L}_s(x, u, w, \lambda) = 0 \quad (1)$$

where $\mathcal{L}_s$ is a system operator, t is time, $x(t)$ is the state of the system, $u(t)$ is its input, $w(t)$ is some external or internal disturbance, and λ is a parameter that defines the system. Each one of these quantities belongs to a suitable set or vector space and there are a large number of possibilities. For example $\mathcal{L}_s$ may be an algebraic, integral, ordinary or partial differential operator, while x may be a finite-dimensional vector or a function. $u(t)$ is usually a low-dimensional vector.

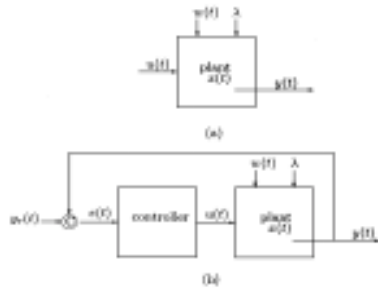


Figure 1: Schematic of a system (a) without and (b) with comparator C and controller.

In general, the output of the system $y(t)$ is different from its state $x(t)$. For example, x may be a spatial temperature distribution, while y are the readings of one or more temperature measurement devices at a finite number of locations. The relation between the two may be written as

$$\mathcal{L}_o(y, x, u, w, \lambda) = 0, \quad (2)$$

where $\mathcal{L}_o$ is the output operator.

The system is *single-input single-output* (SISO) if both u and y are scalars. A system is said to be *controllable* if it can be taken from one specific state to another within a prescribed time with the help of a suitable input. It is *output controllable* if the same can be done to the output. It is important to point out that output controllability does not imply system controllability. In fact, in practice for many thermal systems the former is all that is required; it has been reported that most industrial plants are controlled quite satisfactorily though they are not system controllable [37]. All possible values of the output constitute the *reachable* set. A system is said to be *observable* if its state x can be uniquely determined from the input u and output y alone. The *stability* of a system is a property that leads to a bounded output if the input is also bounded.

2.2 Systems with control

The objective of control is to have a given output $y = y_r(t)$, where the *reference* or *set* value y_r is prescribed. The problem is called *regulation* if y_r is a constant, and *tracking* if it is function of time.

In *open-loop* control the input is selected to give the desired output without using any information from the output side; that is one would have to determine $u(t)$ such that $y = y_r(t)$ using the mathematical model of the system alone. This is a *passive* method of control that is used in many thermal systems. It will work if the behavior of the system is exactly predictable, if precise output control is not required, or if the output of the system is not very sensitive to the input. If the changes desired in the output are very slow then manual control can be carried out, and that is also frequently done. A self-controlling approach that is sometimes useful is to design the system in such a way that any disturbance will bring the output back to the desired value; the output in effect is then insensitive to changes in input or disturbances.

Open-loop control is not usually effective for many systems. For thermal systems contributing factors are the uncertainties in the mathematical model of the plant and the presence of unpredictable disturbances. Internal disturbances may be noise in the measuring or actuating devices or a change in surface heat transfer characteristics due to fouling, while external ones may be a change in the environmental temperature. For these cases *closed-loop* control is an appropriate alternative. This is done using *feedback* from the output, as measured by a *sensor*, to the input side of the system, as shown in Fig. 1(b); the figure actually shows unit feedback. There is a comparator which determines the error signal $e(t) = e(y_r, y)$, which is usually taken to be

$$e = y_r - y. \quad (3)$$

The key role is played by the *controller* which puts out a signal that is used to move an *actuator* in the plant.

Sensors that are commonly used are temperature-measuring devices such as thermocouples, resistance thermometers or thermistors. The actuator can be a fan or a pump if the flow rate is to be changed, or a heater if the heating rate is the appropriate variable. The controller itself is either entirely mechanical if the system is not very complex, or is a digital processor with appropriate software. In any case, it receives the error in the output $e(t)$ and puts out an appropriate control input $u(t)$ that leads to the desired operation of the plant.

The control process can be written as

$$\mathcal{L}_c(u, e, \lambda) = 0 \quad (4)$$

where $\mathcal{L}_c$ is a control operator. The controller designer has to propose a suitable $\mathcal{L}_c$, and then Eqs. (1) - (4) form a set of equations in the unknowns $x(t)$, $y(t)$ and $u(t)$. Choice of a control strategy defines $\mathcal{L}_o$ and many different methodologies are used in thermal systems. It is common to use *on-off* (or bang-bang, relay, etc.) control. This is usually used in heating or cooling systems in which the heat coming in or going out is reduced to zero when a predetermined temperature is reached and set at a constant value at another temperature. Another method is Proportional-Integral-Derivative (PID) control [38] in which

$$u = K_p e(t) + K_i \int_0^t e(s) ds + K_d \frac{de}{dt} . \quad (5)$$

3 LINEAR SYSTEMS THEORY

The term *classical* control is often used to refer to theory derived on the basis of Laplace transforms. Since this is exclusively for linear systems, we will be using the so-called *modern* control or *state-space* analysis which is based on dynamical systems, mainly because it can be extended to nonlinear systems. Where they overlap, the issue is only one of preference since the results are identical. Control theory can be developed for different linear operators, and some of these are outlined below.

3.1 Ordinary differential equations

Much is known about a linear differential system in which Eqs. (1) and (2) take the form

$$\frac{dx}{dt} = Ax + Bu , \quad (6)$$

$$y = Cx + Du , \quad (7)$$

where $x \in \mathbb{R}^n$, $u \in \mathbb{R}^m$, $y \in \mathbb{R}^p$, $A \in \mathbb{R}^{n \times n}$, $B \in \mathbb{R}^{n \times m}$, $C \in \mathbb{R}^{p \times n}$, $D \in \mathbb{R}^{p \times m}$. x , u and y are vectors of different lengths and A , B , C , and D are matrices of suitable sizes. Though A , B , C , and D can be functions of time in general, here they will be treated as constants.

The solution of Eq. (6) is

$$x(t) = e^{A(t-t_0)} x(t_0) + \int_{t_0}^t e^{A(t-s)} Bu(s) ds , \quad (8)$$

where the exponential matrix is defined as

$$e^{At} = I + At + \frac{1}{2!} A^2 t^2 + \frac{1}{3!} A^3 t^3 + \dots ,$$

with I being the identity matrix. From Eq. (7), we get

$$y(t) = C \left[e^{At} x(t_0) + \int_{t_0}^t e^{A(t-s)} Bu(s) ds \right] + Du . \quad (9)$$

Eqs. (8) and (9) define the state $x(t)$ and output $y(t)$ if the input $u(t)$ is given.

It can be shown that for the system governed by Eq. (6), a $u(t)$ can be found to take $x(t)$ from $x(t_0)$ at $t = t_0$ to $x(t_f) = 0$ at $t = t_f$ if and only if the matrix

$$M = [B : AB : A^2 B : \dots : A^{n-1} B] \in \mathbb{R}^{n \times nm} \quad (10)$$

is of rank n . The system is then controllable. For a linear system, controllability from one state to another implies that the system can be taken from *any* state to any other. It must be emphasized that the $u(t)$ that does this is not unique.

Similarly, it can be shown that the output $y(t)$ is controllable if and only if

$$N = [D : CB : CAB : CA^2 B : \dots : CA^{n-1} B] \in \mathbb{R}^{p \times (n+1)m} \quad (11)$$

is of rank p . Also, the state $x(t)$ is observable if and only if the matrix

$$P = [C : CA : CA^2 : \dots : CA^{n-1}]^T \in \mathbb{R}^{pn \times n} \quad (12)$$

is of rank n .

3.2 Algebraic-differential equations

This is a system of equations of the form

$$E \frac{dx}{dt} = Ax + Bu , \quad (13)$$

where the matrix $E \in \mathbb{R}^{n \times n}$ is singular [39]. This is equivalent to a set of equations, some of which are ordinary differential and the rest are algebraic. As a result of this, Eq. (13) cannot be converted into (6) by substitution. The index of the system is the least number of differentiations of the algebraic equations that is needed to get the form of Eq. (6). The system may not be completely controllable since some of the components of x are algebraically related, but it may have restricted or R -controllability [40].

3.3 Partial differential equations

Partial differential equations (PDEs) are an example of infinite-dimensional systems that are very common in thermal applications [41, 42]. *Exact* controllability exists if the function representing the state can be taken from an initial to a final target state, and is *approximate* if it can be taken to a neighborhood of the target [43]. Determination of approximate controllability is usually sufficient for practical purposes.

Consider a system governed by

$$\frac{\partial X}{\partial t} = AX + Bu , \quad (14)$$

with homogeneous boundary and suitable initial conditions, where $\mathcal{A}$ is a bounded semi-group operator [41], and $\mathcal{B}$ is another linear operator. The state $X(\xi, t)$ is a function of spatial coordinates ξ and time t . If $\mathcal{A}$ is self-adjoint, then it has real eigenvalues and a complete orthonormal set of eigenfunctions $\phi_m(\xi)$, with $m = 0, 1, 2, \dots$, which forms a complete spatial basis for X . It is known [42] that the system is approximately state controllable if and only if all the inner products

$$\langle \mathcal{B}, \phi_m \rangle \neq 0 \quad (15)$$

4 NONLINEAR ASPECTS

The following are a few of the issues that arise in the treatment of nonlinear thermal control problems.

4.1 Models

There are no general mathematical models for thermal systems, but one that can be used is a generalization of Eq. (6) such as

$$\frac{dx}{dt} = f(x, u) \quad (16)$$

where $f : \mathbb{R}^n \times \mathbb{R}^m \mapsto \mathbb{R}^n$. If one is interested in local behavior about an equilibrium state $x = x_0$, $u = 0$, this can be linearized in that neighborhood to give

$$\begin{aligned} \frac{dx}{dt} &= \left. \frac{\partial f}{\partial x} \right|_0 x' + \left. \frac{\partial f}{\partial u} \right|_0 u' \\ &= Ax' + Bu', \end{aligned} \quad (17)$$

where $x = x_0 + x'$ and $u = u'$. The Jacobian matrices $(\partial f / \partial x)_0$ and $(\partial f / \partial u)_0$, are evaluated at the equilibrium point. Eq. (17) has the same form as Eq. (6).

4.2 Controllability and reachability

General theorems for the controllability of nonlinear systems are not available at this point in time. Results obtained from the linearized equations generally do not hold for the nonlinear equations. The reason is that in the nonlinear case one can take a path in state space that travels far from the equilibrium point and then returns close to it. Thus regions of state space that are unreachable with the linearized equations may actually be reachable. In a thermal convection loop it is possible to go from one branch of a bifurcation solution to another in this fashion [44].

4.3 Bounded variables

In practice, due to hardware constraints it is common to have the physical variables confined to certain ranges, so that variables such as x and u in Eqs. (1) and (2), being temperatures, heat rates, flow rates and the like, are bounded. If this happens, even systems locally governed by Eqs. (6) and (7) are now nonlinear since the sum of solutions may fall outside the range in which x exists and thus may not be a valid solution. On the other hand, for a controllable system in which only u is bounded in a neighborhood of zero, x can reach any point in $\mathbb{R}^n$ if the eigenvalues of A have zero or positive real parts, and the origin is reachable if the eigenvalues of A have zero or negative real parts [35].

4.4 Relay and hysteresis

A relay is an element of a system that has an input-output relationship that is not smooth; it may be discontinuous or not possess first or higher-order derivatives. This may be accompanied by hysteresis where the relationship also depends on whether the input is increasing or decreasing. Valves are typical elements in flow systems that have this kind of behavior.

5 SYSTEM IDENTIFICATION

To be able to design appropriate control systems, one needs to have some idea of the dynamic behavior of the thermal system that is being controlled. Mathematical models of these systems can be obtained in two entirely different ways: from first principles using known physical laws, and empirically from the analysis of experimental information (though combinations of the two paths are not only possible but common). The latter is the process of *system identification*, by which a complex system is reduced to mathematical form using experimental data [31, 45, 46]. There are many different ways in which this can be done, the most common being the fitting of parameters to proposed models [47]. In this method, a form of $\mathcal{L}_s$ is assumed with unknown parameter values. Through optimization routines the values of the unknowns are chosen to obtain the best fit of the results of the model with experimental information. Apart from the linear Eq. (6), other models that are used include the following.

- There are many forms based on Eq. (16), one of which is the closed-affine model

$$\frac{dx}{dt} = F_1(x) + F_2(x)u \quad (18)$$

The bilinear equation for which $F_1(x) = Ax$ and $F_2(x) = Nx + b$ is a special case of this.

- Volterra models, like

$$y(t) = y_0(t) + \sum_{i=1}^{\infty} \int_{-\infty}^{\infty} \dots \int_{-\infty}^{\infty} k_i(t; t_1, t_2, t_3, \dots, t_i) u(t_1) \dots u(t_i) dt_1 \dots dt_i \quad (19)$$

for a SISO system, are also used.

- Functional [48], difference [49] or delay [50] equations such as

$$\frac{dx}{dt} = Ax(t-s) + Bu \quad (20)$$

also appear in the modeling of thermal systems.

- Fractional-order derivatives, of which there are several different possible definitions [51-55] can be used in differential models. As an example, the Riemann-Liouville definition of the n th derivative of $f(t)$ is

$${}_a D_t^n f(t) = \frac{1}{\Gamma(m-n+1)} \frac{d^{m+1}}{dt^{m+1}} \int_a^t (t-s)^{m-n} f(s) ds, \quad (21)$$

where a and n are real numbers and m is the largest integer smaller than n .

6 LUMPED SYSTEMS

6.1 Mathematical model

Consider a body that is cooled from its surface by convection to the environment with a constant ambient temperature T_∞ . It also has an internal heat source $Q(t)$ to compensate for this heat loss, and the control objective is to maintain the temperature of the body at a given level by manipulating the heat source. The Biot number for the body is $Bi = hL/k$, where h is the convective heat transfer coefficient, L is a characteristic length dimension of the body, and k is its thermal conductivity. If $Bi < 0.1$, the body can be considered to have a uniform temperature $T(t)$. Under this lumped approximation the energy balance is given by

$$Mc \frac{dT}{dt} + hA_s(T - T_\infty) = Q(t), \quad (22)$$

where M is the mass of the body, c is its specific heat, and A_s is the surface area for convection.

Using Mc/hA_s and $hA_s(T_i - T_\infty)$ as the characteristic time and heat rate, this equation becomes

$$\frac{d\theta}{dt} + \theta = Q(t) \quad (23)$$

Here $\theta = (T - T_\infty)/(T_i - T_\infty)$ where $T(0) = T_i$ so that $\theta(0) = 1$. The other variables are now non-dimensional.

With $x = \theta$, $u = Q$, $n = m = 1$ in Eq. (6), we find from Eq. (10) that $\text{rank}(M) = 1$, so the system is controllable.

Open-loop operation to maintain a given non-dimensional temperature θ_r is easily calculated. Choosing $Q = \theta_r$, it can be shown from the solution of Eq. (23), that is

$$\theta(t) = (1 - \theta_r)e^{-t} + \theta_r, \quad (24)$$

that $\theta \rightarrow \theta_r$ as $t \rightarrow \infty$. In practice, to do this the dimensional parameters hA_s and T_∞ must be exactly known. Since this is usually not the case some form of feedback control is required.

6.2 On-off control

In this simple form of control the heat rate in Eq. (22) has only two values; it is either $Q = Q_0$ or $Q = 0$, depending on whether the heater is on or off, respectively. With the system in its *on* mode, $T \rightarrow T_{\max} = T_\infty + Q_0/hA_s$ as $t \rightarrow \infty$, and in its *off* mode, $T \rightarrow T_{\min} = T_\infty$. Taking the non-dimensional temperature to be

$$\theta = \frac{T - T_{\min}}{T_{\max} - T_{\min}} \quad (25)$$

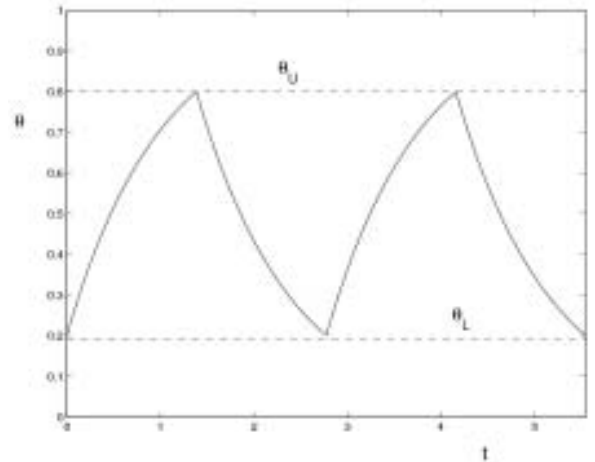


Figure 2: Lumped approximation with on-off control.

the governing equation is

$$\frac{d\theta}{dt} + \theta = \begin{cases} 1 & \text{on} \\ 0 & \text{off} \end{cases}, \quad (26)$$

the solution for which is

$$\theta = \begin{cases} 1 + C_1 e^{-t} & \text{on} \\ C_2 e^{-t} & \text{off} \end{cases} \quad (27)$$

We will assume that the heat source comes on when temperature falls below a value T_L , and goes off when it rises above T_U . These lower and upper bounds are non-dimensionally

$$\theta_L = \frac{T_L - T_{\min}}{T_{\max} - T_{\min}} \quad (28)$$

$$\theta_U = \frac{T_U - T_{\min}}{T_{\max} - T_{\min}} \quad (29)$$

The result of applying this form of control is an oscillatory temperature that looks like that in Fig. 2, the period and amplitude of which can be chosen using suitable parameters. It can be shown that the on and off time periods are

$$t_{on} = \ln \frac{1 - \theta_L}{1 - \theta_U} \quad (30)$$

$$t_{off} = \ln \frac{\theta_U}{\theta_L} \quad (31)$$

respectively. The total period of the oscillation is then

$$t_p = \ln \frac{\theta_U (1 - \theta_L)}{\theta_L (1 - \theta_U)} \quad (32)$$

If we make a small dead-band assumption, we can write

$$\theta_L = \theta_r - \delta \quad (33)$$

$$\theta_U = \theta_r + \delta \quad (34)$$

where $\delta \ll 1$. A Taylor-series expansion gives

$$t_p = 2\delta \left(\frac{1}{\theta_r} + \frac{1}{1 - \theta_r} \right) + \dots \quad (35)$$

The period is thus proportional to the width of the dead band. The frequency of the oscillation increases as its amplitude decreases.

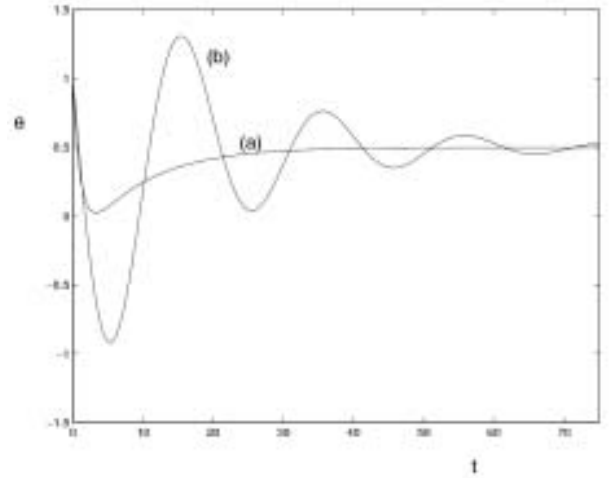


Figure 3: Lumped approximation with PID control; $K_i = K_d = -0.1$, $\theta_r = 0.5$, (a) $K_p = -0.1$, (b) $K_p = -0.9$,

6.3 PID control

The error $e = \theta_r - \theta$ and control input $u = Q$ can be used in Eq. (5), so that the derivative of Eq. (23) gives

$$(K_d + 1) \frac{d^2\theta}{dt^2} + (K_p + 1) \frac{d\theta}{dt} + K_i \theta = K_i \theta_r \quad (36)$$

with the initial conditions

$$\theta = \theta_o \quad \text{at} \quad t = 0 \quad (37)$$

$$\frac{d\theta}{dt} = -\frac{K_p + 1}{K_d + 1} \theta_o + \frac{K_i \theta_r}{K_d + 1} \quad \text{at} \quad t = 0 \quad (38)$$

The response of the closed-loop system can be obtained as a solution. The steady-state for $K_i \neq 0$ is given by $\theta = \theta_r$. It can be appreciated that different choices of the controller constants K_p , K_i , and K_d will give overdamped or underdamped oscillatory or unstable behavior of the system. Fig. 3 shows two examples of closed-loop behavior with different parameter values.

7 SYSTEMS WITH ONE SPATIAL DIMENSION

The lumped approximation in the previous section, valid for $Bi \ll 1$, is frequently not good enough for thermal systems, and the spatial variation of the temperature must be taken into account. The system is then described by PDEs that represent a formidable challenge for control analysis. The simplest examples occur when only one spatial dimension is present.

7.1 Fins

Fig. 4 shows a fin of length L with convection to the surroundings [56]. It is thin and long enough such that the transverse temperature distribution may be neglected. The temperature field is governed by

$$\frac{\partial T}{\partial t} = \alpha \frac{\partial^2 T}{\partial \xi^2} - \zeta (T - T_\infty) \quad (39)$$

where $T(\xi, t)$ is the temperature distribution that represents the state of the system, T_∞ is the temperature of the surroundings, t is time, and ξ is the longitudinal coordinate measured from one end. The thermal diffusivity is α , and $\zeta = hp / \rho c A_c$ where h is the convective heat transfer coefficient, A_c is the constant cross-sectional area of the bar, P is the perimeter of the cross section, ρ is the density, and c is the specic heat. For simplicity it will be assumed that ζ is independent of ξ . The end $\xi = 0$ will be assumed to be adiabatic so that $(\partial T / \partial \xi)(0, t) = 0$.

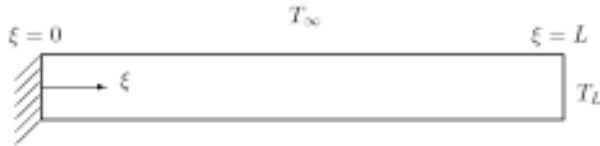


Figure 4: One-dimensional fin with convection.

Since a linear system that is controllable can be taken from any state to any other, we can arbitrarily assume the fin to be initially at a uniform temperature. There are two ways in which the temperature distribution on the bar can be controlled: in *distributed* control¹ the surrounding temperature T_∞ is the control input and in *boundary* control it is the temperature of the other end $T(L, t)$ of the fin.

(a) *Distributed control*: The boundary temperature $T(L, t) = T_L$ is fixed. Using it as a reference temperature and defining $\theta = T - T_L$, Eq. (39) becomes,

$$\frac{\partial \theta}{\partial t} = \alpha \frac{\partial^2 \theta}{\partial \xi^2} - \zeta \theta + \zeta \theta_\infty(t) \quad (40)$$

with the homogeneous boundary and initial conditions $(\partial \theta / \partial \xi)(0, t) = 0$, $\theta(L, t) = 0$, and $\theta(\xi, 0) = 0$.

The operators in Eq. (14) are $\mathcal{A} = \alpha \partial^2 / \partial \xi^2 - \zeta$, $\mathcal{B} = \zeta$, and $u = \theta_\infty$. $\mathcal{A}$ is a self-adjoint operator with the eigenvalues and eigenfunctions

$$\beta_m = -\frac{(2m+1)^2 \pi^2}{4L^2} - \zeta,$$

$$\phi_m = \sqrt{\frac{2}{L}} \cos \frac{(2m+1)\pi \xi}{2L},$$

respectively. Inequality (15) is satisfied for all m , so the system is indeed state controllable. It can be shown that the same problem can also analyzed using a finite-difference approximation [57].

(b) *Boundary control*: Using the constant outside temperature T_∞ as reference and defining $\theta = T - T_\infty$, Eq. (39) becomes

$$\frac{\partial \theta}{\partial t} = \alpha \frac{\partial^2 \theta}{\partial \xi^2} - \zeta \theta, \quad (41)$$

with the initial and boundary conditions $(\partial \theta / \partial \xi)(0, t) = 0$, $\theta(L, t) = T_L(t) - T_\infty$, and $\theta(\xi, 0) = 0$.

To enable a finite-difference approximation [57], the domain $[0, L]$ is divided into n equal parts of size $\Delta \xi$, so that Eq. (41) becomes

$$\frac{d\theta_i}{dt} = \sigma \theta_{i-1} - (2\sigma + \zeta) \theta_i + \sigma \theta_{i+1}, \quad (42)$$

where $\sigma = \alpha / \Delta \xi^2$. The nodes are $i = 1, 2, \dots, n+1$, where $i = 1$ is at the left and $i = n+1$ at the right end of the fin in Fig. 4. With this Eq. (41) can be discretized to take the form of Eq. (6), where x is the vector of unknown θ_i . Thus we find

$$A = \begin{bmatrix} -(2\sigma + \zeta) & 2\sigma & 0 & \cdots & 0 \\ \sigma & -(2\sigma + \zeta) & \sigma & & \vdots \\ 0 & \ddots & \ddots & \ddots & \\ \vdots & & & & \sigma \\ 0 & \cdots & 0 & \sigma & -(2\sigma + \zeta) \end{bmatrix} \in \mathbb{R}^{n \times n}, \quad (43)$$

$$B = [0, \dots, \sigma]^T \in \mathbb{R}^n. \quad (44)$$

¹This term is also used in other senses in control theory.

The boundary conditions have been applied to make A non-singular: at the left end the fin is adiabatic, and at the right end θ_{n+1} is the control input u .

The controllability matrix M is

$$M = \begin{bmatrix} 0 & \cdots & \vdots & 0 & \sigma^n \\ 0 & \cdots & 0 & \sigma^{n-1} & \cdots \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & \sigma^3 & \cdots & \cdots \\ 0 & \sigma^2 & -2\sigma^2(2\sigma+\zeta) & \cdots & \cdots \\ \sigma & -\sigma(2\sigma+\zeta) & \sigma^3+\sigma(2\sigma+\zeta)^2 & \cdots & \cdots \end{bmatrix} \quad (45)$$

The rank of M is n , indicating that the state of the system is also boundary controllable.

7.2 Long duct

The diffusion problem of the previous section does not have advection. Transport of fluids in ducts introduces a delay between the instant the particles of fluid go into the duct and when they come out, which creates a difficulty for outlet temperature control. The literature includes applications to hot-water systems [58,59] and buildings [60, 61]; transport [62] and heater [63, 64] delay and the effect of the length of a duct on delay [65] have also been looked at.

A long duct of constant cross section, schematically shown in Fig. 5 where the flow is driven by a variable-speed pump, illustrates the basic issues [56, 66]. The fluid inlet temperature T_{in} is kept constant, and there is heat loss to the constant ambient temperature T_{∞} through the surface of the duct.

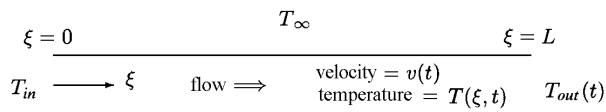


Figure 5: Schematic of duct.

With a one-dimensional approximation, energy conservation gives

$$\frac{\partial T}{\partial t} + v \frac{\partial T}{\partial \xi} + \frac{4h}{\rho c D} (T - T_{\infty}) = 0, \quad (46)$$

with the boundary condition $T(0, t) = T_{in}$, where $T(\xi, t)$ is the fluid temperature, t is time, ξ is the distance along the

duct measured from the entrance, $v(t)$ is the flow velocity, h is the coefficient of heat transfer to the exterior, ρ is the fluid density, c is its specific heat, and D is the hydraulic diameter of the duct. The flow velocity is taken to be always positive, so that the $\xi = 0$ end is always the inlet and $\xi = L$ the outlet, where L is the length of the duct. The temperature of the fluid coming out of the duct is $T_{out}(t)$.

Using the characteristic quantities of L for length, $\rho c D / 4h$ for time, and $hL / \rho c D$ for velocity, the non-dimensional version of Eq. (46) is

$$\frac{\partial \theta}{\partial t} + v \frac{\partial \theta}{\partial \xi} + \theta = 0, \quad (47)$$

where $\theta = (T - T_{\infty}) / (T_{in} - T_{\infty})$, with $\theta(0, t) = 1$. The other variables are now non-dimensional. Knowing $v(t)$, this can be solved to give

$$\theta(\xi, t) = e^{-t} f \left(\xi - \int_0^t v(s) ds \right), \quad (48)$$

where the initial startup interval in which the fluid within the duct is flushed out has been ignored; f is an arbitrary function.

Applying the boundary condition at $\xi = 0$ gives

$$1 = e^{-t} f \left(- \int_0^t v(s) ds \right). \quad (49)$$

The temperature at the outlet of the duct, i.e. at $\xi = 1$, is

$$\theta_{out}(t) = e^{-t} f \left(1 - \int_0^t v(s) ds \right) \quad (50)$$

Eqs. (49) and (50) must be simultaneously solved to get the outlet temperature $\theta_{out}(t)$ in terms of the flow velocity v .

The problem is non-linear if the outlet temperature $T_{out}(t)$ is used to control the flow velocity $v(t)$. The delay between the velocity change and its effect on the outlet temperature can often lead to instability, as it does in other applications [67-70]. Fig. 6 shows a typical result using PID control in which the system is unstable. Shown are the outlet temperature, flow velocity and residence time of the fluid in the duct, all of which ultimately achieve constant amplitude oscillations.

7.3 Heat exchanger model

A cross-flow heat exchanger model, schematically shown in Fig. 7, has been studied using finite differences [56,57,71].

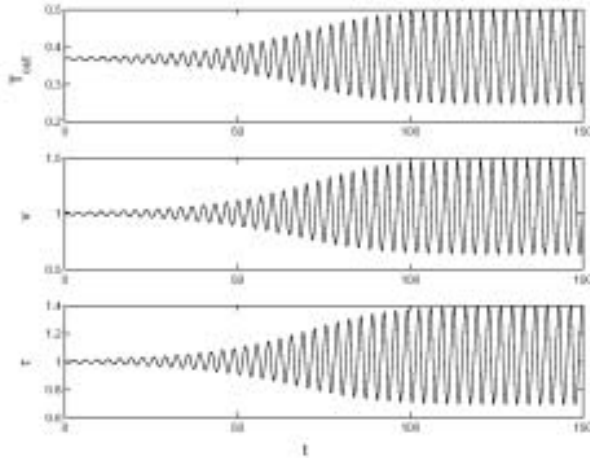


Figure 6: Outlet temperature, velocity and residence time for $K_i = -5$ and $K_p = 2.5$ [56].

Water is the in-tube and air the over-tube fluid in the heat exchanger. This example includes all the conductive, advective and convective effects discussed before. The governing equations on the outside of the tube, in the water, and in the wall of the tube are

$$\frac{\dot{m}_a}{L} c_a (T_{in}^a - T_{out}^a) = h_o 2\pi r_o (T_a - T_t) \quad (51)$$

$$\rho_w c_w \pi r_i^2 \frac{\partial T_w}{\partial t} + \dot{m}_w c_w \frac{\partial T_w}{\partial \xi} = h_i 2\pi r_i (T_t - T_w) \quad (52)$$

$$\rho_t c_t \pi (r_o^2 - r_i^2) \frac{\partial T_t}{\partial t} = k_t \pi (r_o^2 - r_i^2) \frac{\partial^2 T_t}{\partial \xi^2} + 2\pi r_o h_o (T_a - T_t) - 2\pi r_i h_i (T_t - T_w) \quad (53)$$

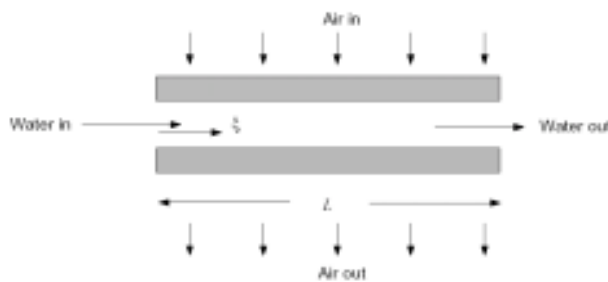


Figure 7: Schematic of single-tube cross-flow heat exchanger.

respectively. L is the length of the tube; $\dot{m}_a(t)$ and $\dot{m}_w(t)$ are the mass flow rates of air and water; T_{in}^a and $T_{out}^a(t)$ are the inlet and outlet air temperatures; $T_a(t)$ is the air temperature surrounding the tube; $T_t(\xi, t)$ and $T_w(\xi, t)$ are the tube-wall

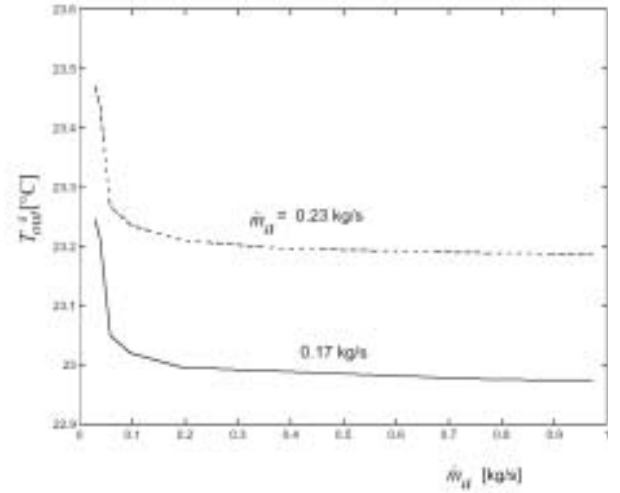


Figure 8: The relation between T_{out}^a and $\dot{m}_w$ for different $\dot{m}_a$ [56].

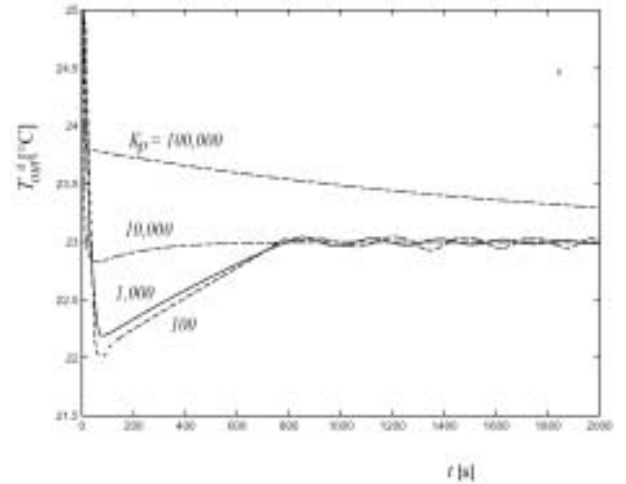


Figure 9: Behavior of T_{out}^a as a function of time for $K_i = 50$ and different K_p [56].

and water temperatures; h_i and h_o the heat transfer coefficients in the inner and outer surfaces of the tube; r_i and r_o are the inner and outer radii of the tube; c_a , c_w and c_t are the specific heats of the air, water and tube material; ρ_w and ρ_t are the water and tube material densities; and k_t is the thermal conductivity of the tube material. In addition, the air temperature is assumed to be $T_a = (T_{in}^a + T_{out}^a)/2$. The boundary and initial conditions are $T_t(0, t) = T_w(0, t) = T_{in}^w$, $T_t(L, t) = T_w(L, t)$, and $T_t(\xi, 0) = T_w(\xi, 0)$. Suitable numerical values were assumed for the computations.

The inlet temperatures T_{in}^a and T_{in}^w , and the flow rates $\dot{m}_a$ and $\dot{m}_w$ can all be used as control inputs to obtain a desired

outlet temperature, T_{out}^a or T_{out}^w . The flow rates present a special difficulty; they appear in nonlinear form in Eqs. (51) and (52), and the outlet temperature is bounded. Fig. 8 shows the steady-state range of values of T_{out}^w that can be achieved on varying $\dot{m}_w$; temperatures outside this range cannot be obtained. It is also seen that the outlet water temperature is hard to control for large water flow rates. As an example of control dynamics, Fig. 9 shows the results of applying PI control on P_w to obtain a given reference temperature $T_{out}^w = 23^\circ\text{C}$.

8 COMPLEX THERMAL SYSTEMS

The previous sections examined systems that were fairly simple in the sense that mathematical models could be written down and their behaviors studied. For most practical thermal systems, this is difficult to do with any degree of precision. In the following, we will look first at thermal components and then combine them in networks.

8.1 Heat exchangers

The most important of the components are heat exchangers, which are generally very complex in that they cannot be realistically computed in real time for control purposes [72-74]. An approach that is becoming popular in these cases is that of artificial neural networks (ANN) [75] for prediction of system behavior both for time-independent [76-79] and time-dependent operation. It is particularly suitable for systems for which experimental information that can be used for training is available. Reviews of artificial neural network applications to thermal engineering [80,81] and other soft control methodologies [82-84] and applications [85] are available.

A stabilized neurocontrol technique for heat exchangers has been described in [86-90]. Fig. 10 shows the test facility in which the experiments were conducted. The objective is to control the outlet air temperature T_{out}^a . Figs. 11 shows the results of using neurocontrol compared with PID; both are effective. Fig. 12 shows the result of a disturbance rejection experiment. The heat exchanger is stabilized at $T_{out}^a = 36^\circ\text{C}$, and then the water flow is shut down between $t = 40\text{ s}$ and $t = 70\text{ s}$; after that the neurocontroller brings the system back to normal operation. A neural network-based controller is able to adapt easily to changing circumstances; in thermal systems this may come from effects such as the presence of fouling over time or from changes in system configuration as could happen in building heating and cooling systems.

8.2 Hydronic networks

The science of networks of all kinds has been put forward as a new emerging science [91]. In the present context this means that a complete understanding of the behavior of components does not necessarily mean that large networks formed out of these components can be modeled and computed in real time for control purposes. Controllability issues of heat exchanger networks are reported in [92]. Mathematical models of the dynamics of a piping network lead to differential-algebraic systems [93]. The momentum equation governing the flow in each pipe is differential, while the conservation of mass condition at each of the junctions is algebraic. Thus, it turns out that only certain flow rates may be controllable, the others being dependent on these.

There are at present many different strategies for the thermal control of networks, and comparative studies based on mathematical models can be carried out. Fig. 13 shows a network in which three specific control strategies can be

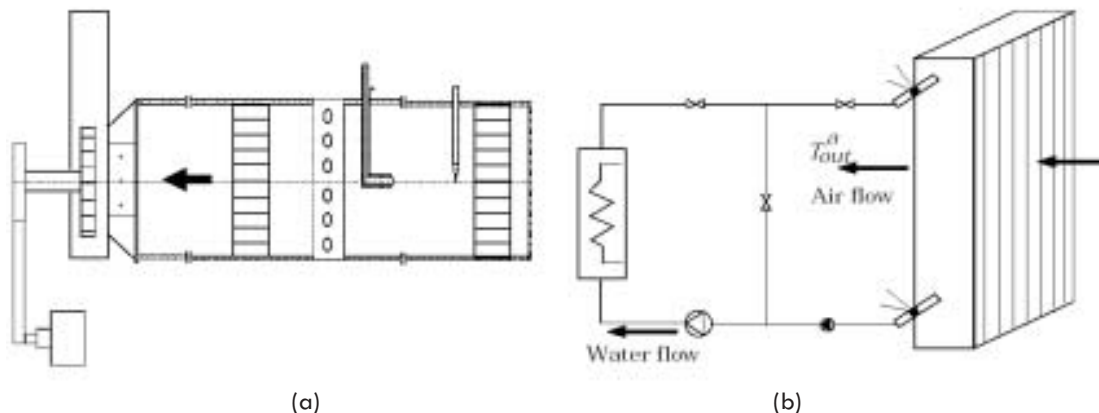


Figure 10: Experimental setup: (a) heat exchanger test facility with wind tunnel and in-draft fan, (b) heat exchanger with water and air flows indicated; T_{out}^a is the air outlet temperature [89].

compared [93, 94]; each control method works differently and are labeled VF, MCF and BT in the following figures (details are in [93]). The network has a primary loop, a secondary loop and a bypass that has the three strategies as special cases. The primary loop includes a chiller, while the secondary has a water-air cooling coil which serves as a thermal load. Integral controllers are used to operate the valves V_α , V_β and V_γ to control the air temperature leaving the cooling coil, $T_a^L(t)$. Figs. 14 and 15 show the dynamic response of $T_a^L(t)$, the leaving water temperatures $T_w^L(t)$, and the bypass pressure difference Δp_{bp} to step changes in the air velocity over the coil. $\alpha(t)$, $\beta(t)$ and $\gamma(t)$ are the respective closing fractions of the valves which change dynamically in response to the error signal. There are some oscillations in all the variables before they settle down to stable, steady values.

Laboratory experiments with a network of water-to-water heat exchangers have been reported in [93-95]; the configuration is shown in Fig. 16. The hot water flow is diminished by changing its controller set point. Figure 17 shows the secondary hot water flow rates q_8 , q_7 and q_4 to the heat exchangers for the three different control strategies. Each curve represents one independent run; that is, water flow to HX_{BT} and HX_{CF} is zero when testing VF, and so on. The system is taken to the nominal operating conditions, and then the hot water flow is decreased by a constant value every 1800 s. The controls drive the system to different operational points while coping with the changes. The input voltages v_7 , v_4 and v_1 that control flow and the hot water temperature at the heat exchanger inlet T_{21} are also shown. It is seen that for certain control parameters, the system is unstable and the variables oscillate in time.

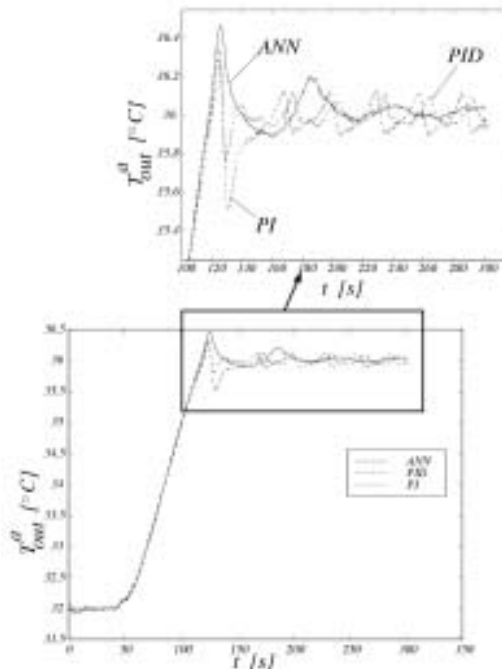


Figure 11: Time-dependent behavior of heat exchanger using ANN, PID and PI control methodologies [89].

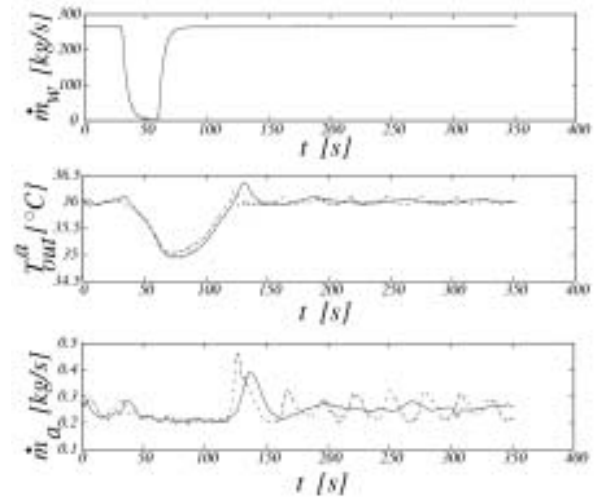


Figure 12: Time-dependent behavior of heat exchanger with neurocontrol for disturbance rejection experiment showing flow rates and air outlet temperature [89].

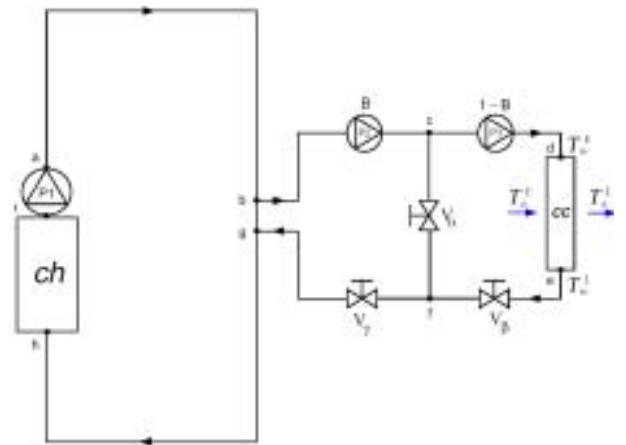


Figure 13: Network used to study control strategies [93].

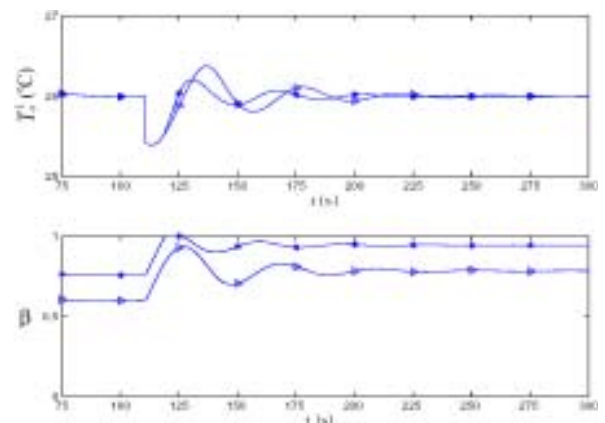


Figure 14: Dynamic response of control system to drop in air velocity, —*— method VF, and —▷— method MCF; $T_{a, set}^L = 26^\circ\text{C}$, $T_a^E = 30^\circ\text{C}$ [93].

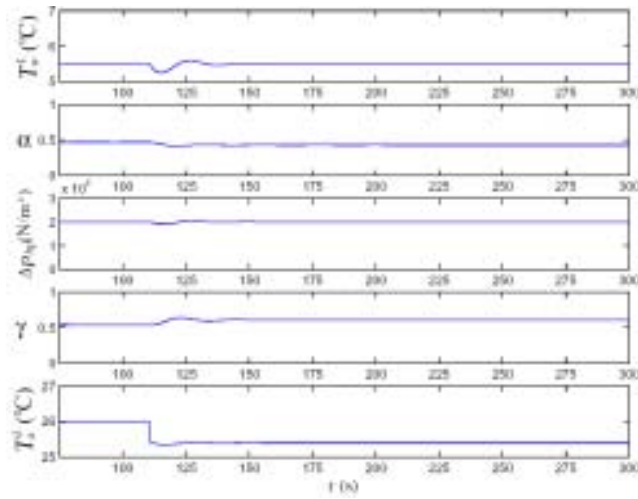


Figure 15: Dynamic response of control system to drop in air velocity with method BT; $T_{w, set}^L = 5.5^\circ \text{C}$, $\Delta p_{sp, set} = 20000 \text{ (N/m}^2\text{)}$; $T_a^E = 30^\circ \text{C}$ [93].

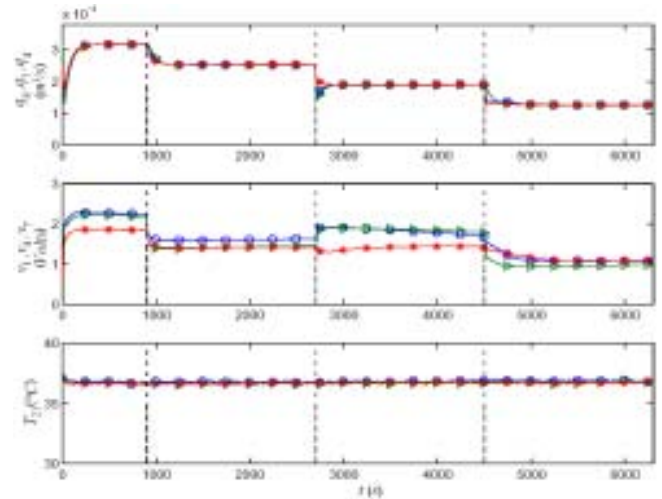


Figure 16: Layout of hydronic network [93].

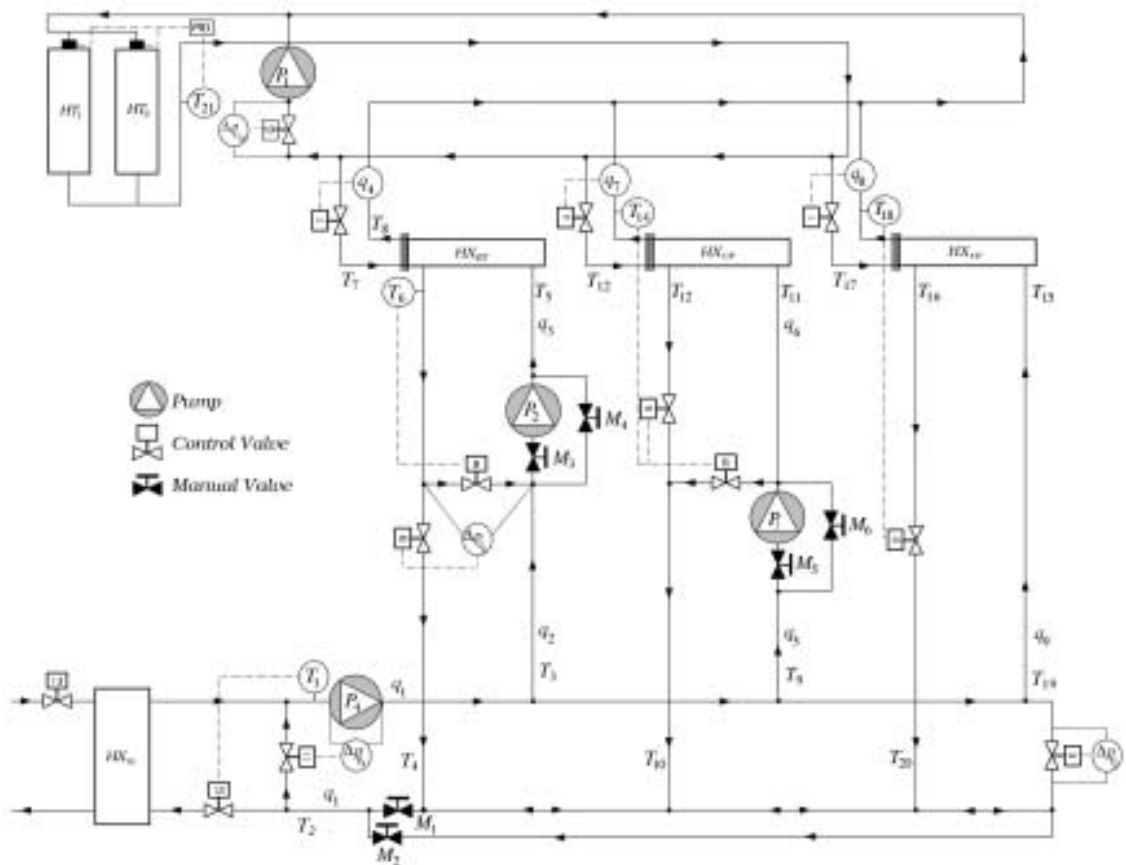


Figure 17: Secondary hot water flows and T_{21} : $\circ$ — BT, $\triangleright$ — CF, $*$ — VF. The dashed vertical lines are instants at which the thermal load is changed [93].

8.3 Other applications

There are a large number of other thermal problems in which control theory has been applied. Agent-based controls have been proposed by complex thermal systems such as in buildings [96], microwave heating [97], thermal radiation [98], and materials processing and manufacturing [99, 100]. Control of convection is an important and active topic; this includes the study of convection loops [101-105], stabilization and control of convection in horizontal fluid layers [106-115], and in porous media [116].

9 CONCLUSIONS

This has been a very brief introduction to the theory of thermal control. The fundamental ideas in this subject are firmly grounded on the mathematics of systems and control theory which should be the starting point. There are, however, a few aspects that are particularly characteristic of thermal systems. Phenomena such as diffusion, convection and advection are common and the systems are usually complex, nonlinear and poorly predictable dynamically. The governing equations cover a wide range of possibilities, from ordinary and partial differential equations to functional and differential-algebraic systems. Furthermore, control theory itself is a vast subject, with specialized branches like optimal [117], robust [118], and stochastic control [119] that are well developed. Many of the tools in these areas find applications in thermal systems.

The study of thermal control will continue to grow from the point of view of fundamentals as well as engineering applications. There are many outstanding problems and issues that need to be addressed. To cite one specific example, networking between a large number of coupled components will become increasingly important; it is known that unexpected synchronization may result even when multiple dynamical systems are coupled weakly [120]. It is hoped that the reader will use this brief overview as a starting point for further study and apply control theory in other thermal applications.

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